

A Research Analysis on Security Challenges and Risk Mitigation Strategies in Cloud Computing Environments - AWS

Submitted By

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2024-2025

Project Overview

- ☐ Cloud computing security analysis project
 - ☐ AWS-based practical implementation
 - ☐ Identification of cloud security threats
 - ☐ Implementation of risk mitigation strategies
 - ☐ Monitoring and security analysis using AWS services
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Objectives of the Project

- ☐ Analyze major cloud security threats
 - ☐ Implement AWS cloud security services
 - ☐ Monitor cloud activities and logs
 - ☐ Apply risk mitigation strategies
 - ☐ Evaluate security frameworks
 - ☐ Study future cloud security trends
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Problem Statement

❑ Organizations face multiple cloud security risks such as:

- Data breaches
 - Public cloud storage exposure
 - Weak IAM permissions
 - Unauthorized access
 - Misconfiguration
 - Lack of monitoring and logging
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Technologies & Tools Used

☐ AWS Services

- Amazon S3
- AWS IAM
- AWS CloudTrail
- Amazon GuardDuty
- AWS Config

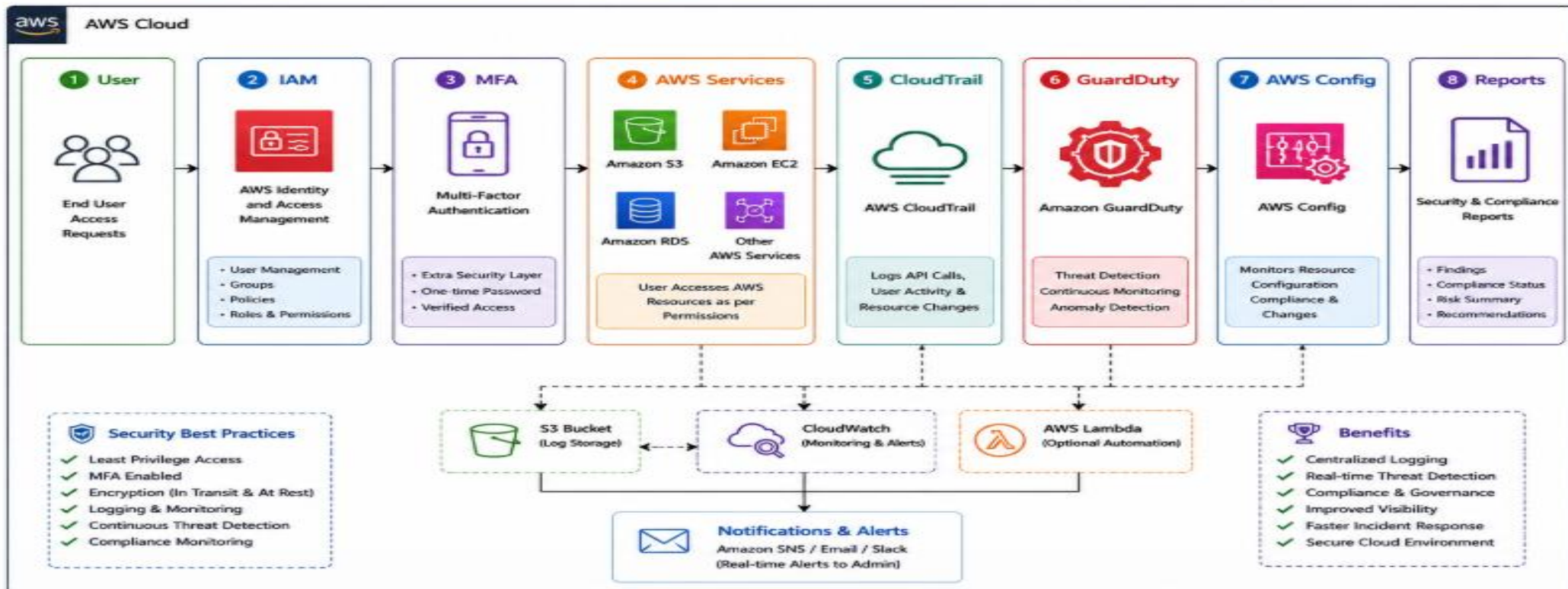
☐ Other Tools

- MFA Authentication
 - Monitoring & Logging
 - Security Analysis
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System Architecture

AWS Cloud Security Architecture

Workflow: User → IAM → MFA → AWS Services → CloudTrail → GuardDuty → AWS Config → Reports



Security Threats Identified

☐ Security Threats

- Public S3 Bucket Exposure
 - Excessive IAM Permissions
 - Weak Authentication
 - Misconfiguration
 - Unauthorized Access
 - Lack of Continuous Monitoring
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Security Frameworks Implemented

☐ Security Frameworks

- Shared Responsibility Model
 - IAM Access Control
 - MFA Authentication
 - Encryption
 - Cloud Monitoring
 - Zero Trust Concept
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AWS Implementation – Amazon S3

☐ Amazon S3 Security Implementation

☐ Implemented:

- S3 Bucket Creation
 - Public Access Block
 - Encryption Enabled
 - Versioning Enabled
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AWS Implementation – IAM

☐ IAM Security Implementation

☐ Implemented:

- IAM User Creation
 - Role-Based Access
 - Least Privilege Access
 - MFA Enabled
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AWS Monitoring Services

☐ Monitoring & Threat Detection

☐ CloudTrail

- Activity logging
- Event monitoring

☐ GuardDuty

- Threat detection
- Suspicious activity monitoring

☐ AWS Config

- Compliance monitoring
 - Configuration tracking
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Data Analysis

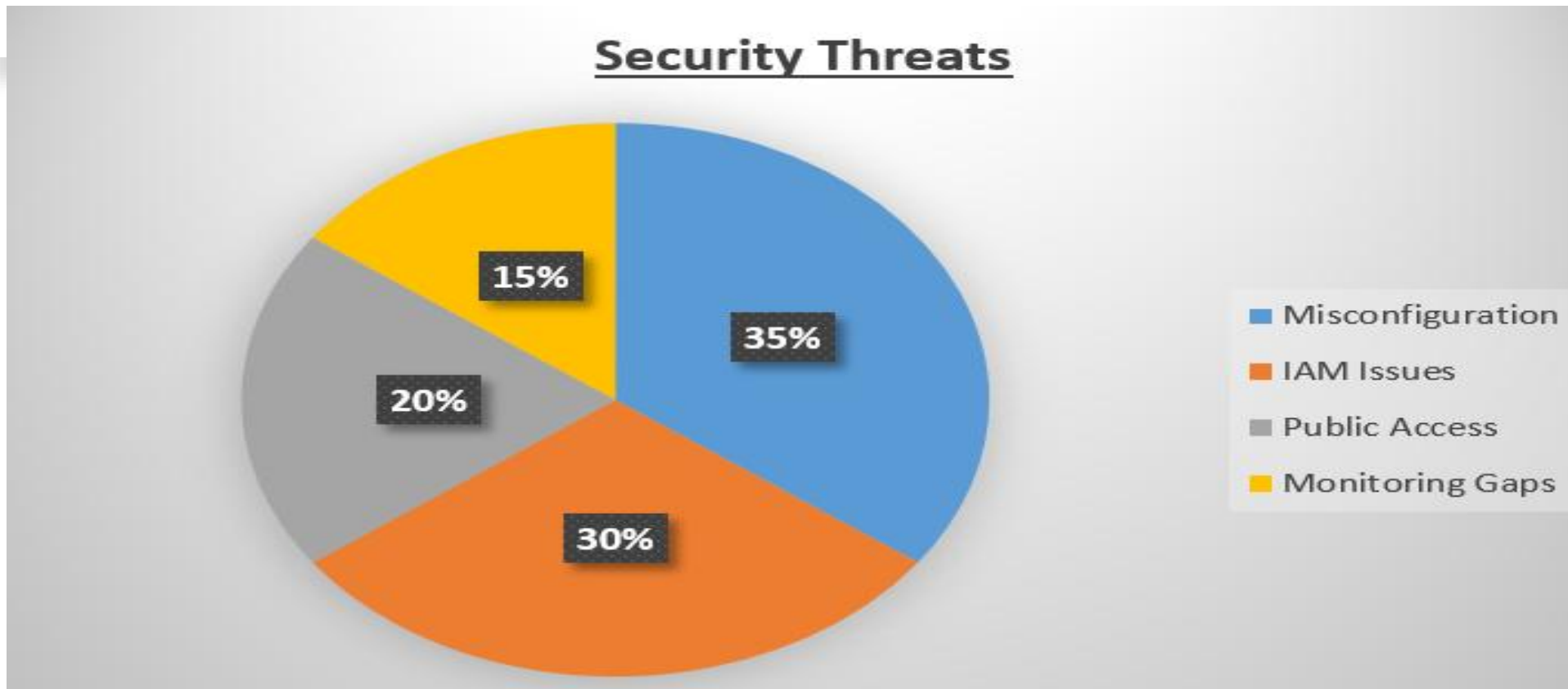
☐ Security Analysis

☐ Threat Distribution

- Misconfiguration
 - IAM Risks
 - Public Access Risks
 - Monitoring Gaps
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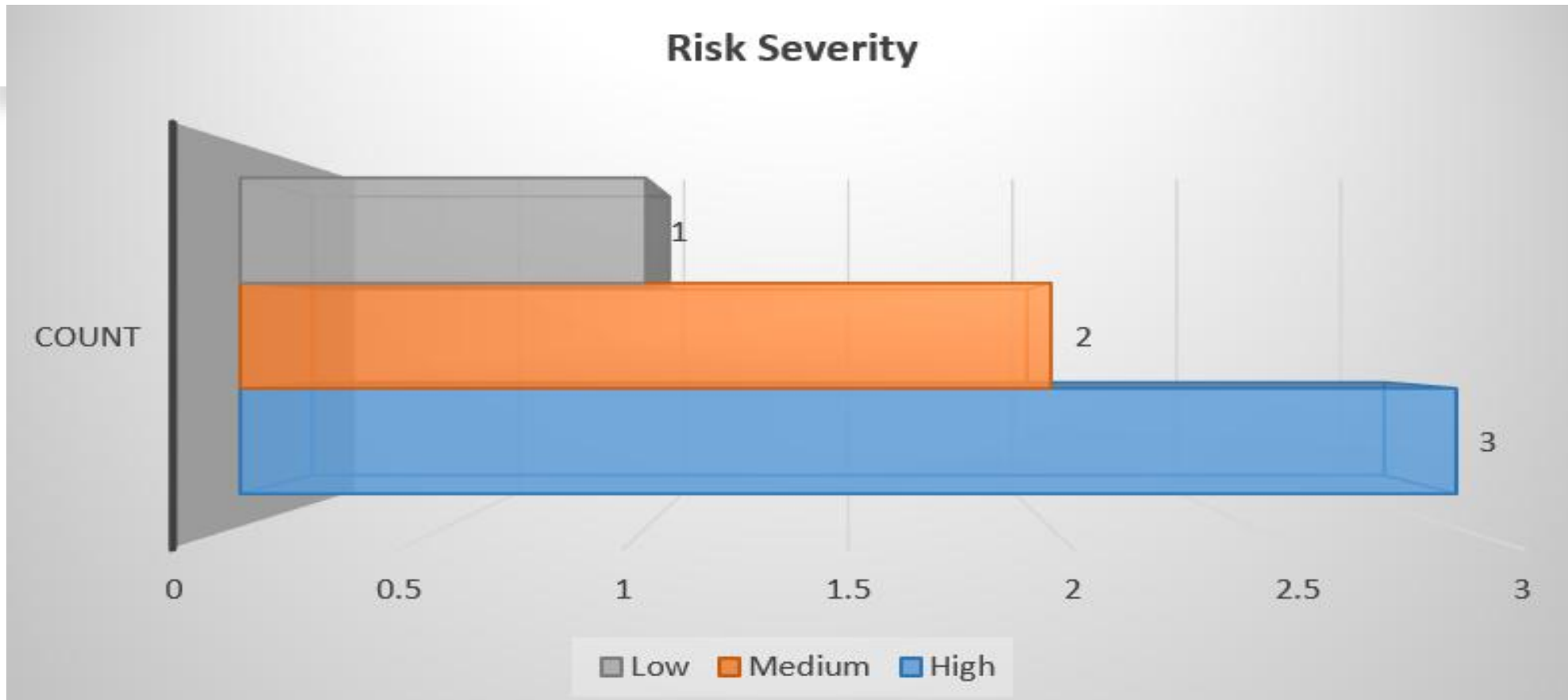
Pie Chart- Security Threats

Distribution of Security Threats by my own analysis:-



Bar Garph – Risk Severity

Risk Severity by my own analysis:-



Testing

Testing Performed:-

Test	Result
MFA Validation	Pass
Public Access Block	Pass
<u>CloudTrail</u> Logging	Pass
Threat Monitoring	Pass

Results & Findings

□ Key Findings

- Misconfiguration is a major cloud security risk
 - IAM controls improve access security
 - MFA reduces account hijacking risks
 - Continuous monitoring improves threat visibility
 - Encryption protects sensitive cloud data
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Future Scope

❑ Future Enhancements

- AI-based threat detection
 - SIEM integration
 - Multi-cloud monitoring
 - Automated remediation
 - Advanced Zero Trust implementation
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Conclusion

- ☐ Successfully analyzed cloud security threats
 - ☐ Implemented AWS security controls
 - ☐ Applied risk mitigation strategies
 - ☐ Improved monitoring and cloud security posture
 - ☐ Achieved project objectives successfully
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Thank You!!!
